**SIMATS ENGINEERING**
**ASSESSMENT TOOL 1**

**COURSE NAME: Fundamentals of Data Science** **COURSE CODE: DSA0402**

---

**COURSE OUTCOME COVERED**

CO2: Apply data pre-processing and visualization techniques using Python libraries for effective data handling (BL4)

---

**QUESTIONS: 20** **Total Marks:100**

Annexure A

**MCQ**

---

***Code of Conduct:***

*I **Jannathul Firdaus. N (192573025)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
*I understand that any violation of academic integrity rules will result in disciplinary action.*

***Signature of the student:*** **Jannathul Firdaus. N**

## RUBRICS FOR CALCULATION

| Criteria | Weightage | Level 4 – Exemplary (5 Marks) | Level 3 – Proficient (4 Marks) | Level 2 – Developing (2–3 Marks) | Level 1 – Beginning (0–1 Mark) | Level | Marks |
|---|---|---|---|---|---|---|---|
| **Conceptual Understanding** | **25** | Demonstrates clear and complete understanding of the concept tested in the MCQ | Shows good understanding with minor conceptual gaps | Shows basic understanding with some misconceptions | Shows poor or incorrect understanding of the concept | | |
| **Correct Option Selection** | **30** | Selects the correct answer accurately without confusion | Selects the correct answer with slight uncertainty | Selects a partially relevant or close option but not fully correct | Selects an incorrect or irrelevant option | | |
| **Analytical Reasoning** | **20** | Clearly analyzes the question and identifies the logic behind the correct option | Provides reasonable analysis with minor errors | Shows limited reasoning or incomplete analysis | Provides no valid reasoning or guesses the answer | | |
| **Elimination of Wrong Options** | **15** | Correctly identifies why all incorrect options are wrong | Identifies most incorrect options with minor explanation gaps | Identifies only some wrong options with weak justification | Does not identify or explain incorrect options | | |
| **Clarity and Response Presentation** | **10** | Response is clear, concise, and well-organized | Response is understandable with minor clarity issues | Response is somewhat unclear or poorly structured | Response is confusing, incomplete, or missing | | |

1. A CSV file is imported into pandas, but the first row of actual data is wrongly treated as column names. What is the best correction.

Use header = None while reading the file.

2. A dataset has 25 columns but only Age, Salary and department are required for analysis. Why should only these columns be selected?

To reduce unnecessary data processing.

3. A manager wants employees from IT department with salary above ₹ 60,000 which command is correct?

df [(df ['department'] == 'IT') & (df ['salary'] > 60000)]

4. A student data set is filtered using:

df [df ['Marks'] >= 50] what does this operation return?

Student who scored 50 or above.

5. A dataset has 40 % missing values in one column what is the most analytical concern before filling missing value?

whether missingness may affect the reliability & analysis.

6. Mean imputation is most suitable when.
The numerical data is approximately symmetric

7. A salary column contains a few very high executive salaries. Some salary values are missing. Which imputation method is more suitable. Median.

8. City column contains missing values. Which method is usually suitable?
Replace missing value with the most frequency city

9. A dataset has 10,000 rows and only 12 rows contains missing values. What is a resonable approach? Drop the 12 is complete rows.

10. A company wants to identify the top 5 highest paid employees. Which sequence is most appropriate?
Sort salary in desending order and select 1st 5 rows

11. A School wants to sort students first by class and then by marks in desending order within each class. Which command is correct?
df.sort_values(['class','marks'], ascending = [True, False])

12. A Sales dataset contains Regions, Product and Sales. To find total Sales for each region and product combination which Operation is best?

df.groupby [('Region', 'product'])['Sales'].sum()

13. A college wants to compare average marks department-wise. Which result best supports this analysis? Mean marks grouped by department.

14. A company wants department-wise average Salary and maximum salary. Which command is Suitable?

df.groupby('Department')['Salary'].agg(['mean', 'max'])

15. Two students have the same mark. What does ranking help identify?

It assigns relative positions while handling ties based on rank method.

16. A company wants to rank employees by salary within each department. Which command is correct.

df['Rank'] = df.groupby('Department')['Salary']
.rank(ascending = False)

17. A researcher wants to compare average marks of 5 departments. Which visualation is most suitable? Bar chart.

18. A company records monthly profit for 24 months. which chart gives the best analytical understanding of growth or decline? line chart.

19. A dataset contains Advertising Budget and Sales Revenue. which visualization is most suitable to analyze whether higher advertisement spending increase sales? Scatter plot.

20. A data set contains employee salaries and the analyst suspects some extremely high salaries. which visualization is most useful? Box plot.